# UNIQA CRO TEAM

## ROBOT DESIGN

### DISTANCE SENSORS:

- THE ROBOT IS EQUIPPED WITH 8 DISTANCE SENSORS PLACED AROUND THE CHASSIS TO DETECT NEARBY WALLS AND OBSTACLES FROM MULTIPLE DIRECTIONS. THESE SENSORS ARE USED FOR WALL FOLLOWING, OBSTACLE AVOIDANCE, CORRIDOR ALIGNMENT, AND MAINTAINING SAFE DISTANCES WHILE NAVIGATING THE MAZE. WE SELECTED MULTIPLE DISTANCE SENSORS BECAUSE THEY PROVIDE FAST AND RELIABLE SHORT-RANGE DETECTION WITH LOW PROCESSING REQUIREMENTS. COMPARED TO RELYING ONLY ON CAMERAS OR LIDAR, THE DISTANCE SENSORS ALLOW THE ROBOT TO REACT MORE QUICKLY TO NEARBY OBSTACLES AND IMPROVE NAVIGATION STABILITY IN TIGHT SPACES.

### GPS

- THE GPS SENSOR IS USED FOR TRACKING THE ROBOT'S POSITION INSIDE THE SIMULATION ENVIRONMENT AND ASSISTING WITH NAVIGATION AND MAPPING. IT HELPS THE ROBOT DETERMINE ITS COORDINATES AND IMPROVES MOVEMENT PRECISION DURING AUTONOMOUS OPERATION. GPS WAS SELECTED BECAUSE IT PROVIDES STABLE POSITIONAL INFORMATION THAT CAN BE COMBINED WITH OTHER SENSORS FOR MORE ACCURATE LOCALIZATION. COMPARED TO RELYING ONLY ON WHEEL MOVEMENT CALCULATIONS, GPS HELPS REDUCE ACCUMULATED NAVIGATION ERRORS OVER TIME.

### COLOUR SENSOR

- THE COLOUR SENSOR IS USED FOR DETECTING IMPORTANT FLOOR MARKINGS AND RESCUE AREA INDICATORS WITHIN THE SIMULATION ENVIRONMENT. IT ASSISTS THE ROBOT IN IDENTIFYING SPECIAL ZONES AND ADAPTING ITS BEHAVIOR ACCORDINGLY. THIS SENSOR WAS SELECTED BECAUSE IT PROVIDES FAST AND RELIABLE SURFACE RECOGNITION WITH MINIMAL COMPUTATIONAL REQUIREMENTS. COMPARED TO CAMERA-BASED FLOOR ANALYSIS, THE COLOUR SENSOR OFFERS SIMPLER IMPLEMENTATION AND FASTER RESPONSE TIMES FOR SPECIFIC TASKS.

### LIDAR

- THE ROBOT IS EQUIPPED WITH A LIDAR SENSOR USED FOR DETAILED ENVIRONMENTAL SCANNING AND MAP GENERATION. LIDAR ALLOWS THE ROBOT TO MEASURE DISTANCES ACROSS A WIDE FIELD OF VIEW AND CREATE AN ACCURATE REPRESENTATION OF THE MAZE. LIDAR WAS SELECTED BECAUSE IT PROVIDES HIGHLY ACCURATE DISTANCE MEASUREMENTS AND SIGNIFICANTLY IMPROVES NAVIGATION RELIABILITY AND PATHFINDING PERFORMANCE. COMPARED TO USING ONLY SHORT-RANGE SENSORS, LIDAR ENABLES BETTER MAPPING OF LARGER AREAS AND IMPROVES AUTONOMOUS DECISION-MAKING.

### CAMERAS:

- THE ROBOT USES 2 CAMERAS CONNECTED TO AN AI-BASED COMPUTER VISION SYSTEM TRAINED FOR VICTIM DETECTION. THE CAMERAS ARE USED TO RECOGNIZE VICTIMS, ANALYZE THE ENVIRONMENT, AND SUPPORT AUTONOMOUS RESCUE OPERATIONS. TWO CAMERAS WERE SELECTED INSTEAD OF A SINGLE CAMERA TO INCREASE THE FIELD OF VIEW AND REDUCE BLIND SPOTS DURING NAVIGATION. CAMERAS COMBINED WITH ARTIFICIAL INTELLIGENCE PROVIDE MORE FLEXIBLE AND ACCURATE VICTIM DETECTION COMPARED TO SIMPLER COLOR-BASED OR SINGLE-SENSOR SOLUTIONS.

### INERTIAL UNIT

- THE ROBOT USES AN INERTIAL UNIT (IMU) TO MEASURE ROTATION, ORIENTATION, AND ACCELERATION. THIS SENSOR HELPS MAINTAIN STABLE MOVEMENT, ACCURATE TURNING ANGLES, AND ORIENTATION AWARENESS DURING NAVIGATION. THE IMU WAS CHOSEN BECAUSE IT IMPROVES TURNING PRECISION AND PROVIDES RELIABLE ORIENTATION DATA DURING MOVEMENT. COMPARED TO USING ONLY WHEEL CALCULATIONS, THE INERTIAL UNIT ALLOWS MORE ACCURATE CONTROL WHEN NAVIGATING COMPLEX RESCUE PATHS.

### SENSOR SELECTION STRATEGY

- THE ROBOT WAS DESIGNED USING A COMBINATION OF COMPLEMENTARY SENSORS INSTEAD OF RELYING ON A SINGLE SENSING SYSTEM. EACH SENSOR WAS SELECTED BASED ON ITS STRENGTHS IN ACCURACY, RESPONSE SPEED, RELIABILITY, AND COMPUTATIONAL EFFICIENCY. DISTANCE SENSORS PROVIDE FAST LOCAL OBSTACLE DETECTION, LIDAR ENABLES ACCURATE MAPPING AND PATH PLANNING, CAMERAS ALLOW AI-BASED VICTIM RECOGNITION, WHILE GPS AND THE INERTIAL UNIT IMPROVE LOCALIZATION AND MOVEMENT PRECISION. COMBINING THESE TECHNOLOGIES INCREASES RELIABILITY AND REDUCES THE WEAKNESSES OF INDIVIDUAL SENSORS, RESULTING IN MORE STABLE AUTONOMOUS NAVIGATION THROUGHOUT THE RESCUE SIMULATION ENVIRONMENT.

### ABOUT CROATIA

- CROATIA IS A MEDITERRANEAN COUNTRY KNOWN FOR ITS BEAUTIFUL ADRIATIC COAST, CRYSTAL-CLEAR SEA, HISTORIC CITIES AND RICH CULTURAL HERITAGE. THE COUNTRY IS HOME TO MORE THAN A THOUSAND ISLANDS, NUMEROUS NATIONAL PARKS AND CITIES SUCH AS DUBROVNIK, SPLIT AND ZAGREB. CROATIA IS ALSO RECOGNIZED WORLDWIDE FOR FOOTBALL, WITH THE NATIONAL TEAM ACHIEVING MAJOR INTERNATIONAL SUCCESS, INCLUDING REACHING THE 2018 FIFA WORLD CUP FINAL AND WINNING BRONZE MEDALS IN 1998 AND 2022.

## SOFTWARE

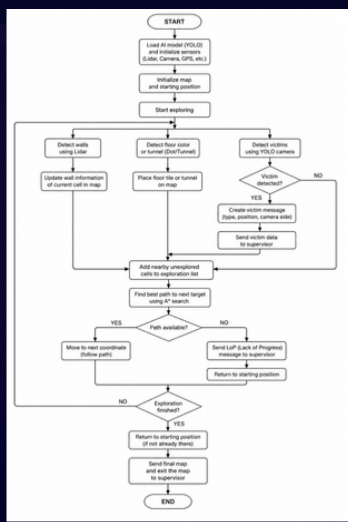
- THE ROBOT SOFTWARE WAS DEVELOPED IN PYTHON USING STANDARD PYTHON LIBRARIES TOGETHER WITH OPENCV, NUMPY, PYTORCH AND THE YOLO AI MODEL. THE SYSTEM RUNS IN THE WEBOTS SIMULATION ENVIRONMENT AND COMMUNICATES DIRECTLY WITH THE ROBOT'S SENSORS, CAMERAS, LIDAR AND MOTORS.
- THE SOFTWARE IS DIVIDED INTO TWO MAIN PARTS:
  - NAVIGATION AND MAPPING SYSTEM
  - VICTIM AND HAZARD DETECTION SYSTEM

### NAVIGATION AND MAPPING

- THE ROBOT AUTONOMOUSLY EXPLORES THE MAZE BY PRIORITIZING UNEXPLORED AREAS AND COMPLETELY EXPLORING ONE ROOM BEFORE MOVING TO THE NEXT ONE.
- THE NAVIGATION SYSTEM USES: LIDAR, DISTANCE SENSORS, GPS, INERTIAL SENSORS, COLOR SENSORS
- THE MAZE IS REPRESENTED AS A GRID-BASED MAP MADE OF INDIVIDUAL CELLS. EACH CELL STORES INFORMATION ABOUT: WALLS, HOLES, SWAMPS, CHECKPOINTS, TUNNELS, VICTIMS, EXPLORED AND UNEXPLORED AREAS
- THE ROBOT CONTINUOUSLY UPDATES THE MAP DURING RUNTIME AND PRINTS A LIVE CONSOLE VISUALIZATION OF THE LABYRINTH FOR EASIER DEBUGGING AND MONITORING.
- THE EXPLORATION ALGORITHM CONSTANTLY SEARCHES FOR THE CLOSEST SAFE UNEXPLORED CELL AND REMOVES UNREACHABLE OR DANGEROUS POSITIONS FROM THE EXPLORATION LIST.

### VICTIM AND HAZARD DETECTION

- WHEN THE ROBOT DETECTS A POSSIBLE VICTIM OR HAZARD, IT CAPTURES AN IMAGE AND SENDS IT TO THE AI RECOGNITION SYSTEM.
- THE SOFTWARE COMBINES TWO DETECTION METHODS:
- MATHEMATICAL DETECTION
- CIRCULAR HAZARDS (C/F/O/P) ARE DETECTED USING HSV COLOR ANALYSIS.
- THE PROGRAM SAMPLES MULTIPLE COLORED RINGS, CONVERTS COLORS INTO NUMERICAL VALUES AND CALCULATES THE FINAL RESULT MATHEMATICALLY.
- AI DETECTION
- LETTER VICTIMS (H/S/U) ARE RECOGNIZED USING A YOLO NEURAL NETWORK MODEL TRAINED FOR VICTIM CLASSIFICATION.
- THE SOFTWARE ALSO INCLUDES: CONFIDENCE FILTERING, FAKE VICTIM REJECTION, SIZE AND RATIO VALIDATION, COOLDOWN PROTECTION TO AVOID DUPLICATE VICTIM REPORTS




## OUR TEAM

### LOVRO ŽIGMAN

TEAM CAPTAIN, ROBOT DESIGNER AND RESPONSIBLE FOR ROBOT NAVIGATION AND AUTONOMOUS MOVEMENT THROUGH THE RESCUE SIMULATION MAZE. DEVELOPS AND OPTIMIZES PATHFINDING ALGORITHMS, MOVEMENT LOGIC, AND DECISION-MAKING SYSTEMS THAT ALLOW THE ROBOT TO EFFICIENTLY NAVIGATE RESCUE ENVIRONMENTS. COORDINATES TEAM ACTIVITIES AND ASSISTS WITH TESTING AND DEBUGGING.

### LUKA KOLARIĆ

TEAM CO-CAPTAIN AND AI DEVELOPER RESPONSIBLE FOR CAMERA INTEGRATION AND TRAINING THE ARTIFICIAL INTELLIGENCE MODEL USED FOR VICTIM DETECTION. WORKS ON COMPUTER VISION SYSTEMS, IMAGE PROCESSING, AND OPTIMIZATION OF DETECTION ACCURACY WHILE ALSO CONTRIBUTING TO SOFTWARE TESTING AND ROBOT PERFORMANCE ANALYSIS.

### ORSAT KRALJ

SOFTWARE SUPPORT MEMBER AND POSTER DESIGNER RESPONSIBLE FOR SIMULATION TESTING, CALIBRATION, AND VISUAL PRESENTATION MATERIALS. ASSISTS IN PROGRAMMING AND DEBUGGING ROBOT BEHAVIOR IN DIFFERENT RESCUE SCENARIOS, ANALYZES SYSTEM PERFORMANCE, AND CONTRIBUTES TO DOCUMENTATION AND TEAM PRESENTATION PREPARATION.

## PAST SUCCESSES

AS A MEMBER OF TEAM HDR LOKOMOTIVA, LOVRO ŽIGMAN ACHIEVED 1ST PLACE IN SUPERTEAMS AND 2ND PLACE INDIVIDUAL AT 2025 EUROPEAN RCJ COMPETITION IN BARI IN THE SOCCER LIGHTWEIGHT ENTRY CATEGORY.

AS A MEMBER OF KICKTRON Z, LUKA KOLARIĆ ACHIEVED 3RDPLACE IN SUPERTEAMS AT 2024 WORLD RCJ COMPETITION IN THE NETHERLANDS IN SOCCER OPEN CATEGORY. HE ALSO ACHIVED 3RD PLACE AT ASIA PACIFIC 2025. COMPETITION IN RCAP COSPACE RESCUE U19 CATEGORY. AT 2025 EUROPEAN RCJ COMPETITION IN BARI HE WON 3RD PLACE IN RESCUE LINE CATEGORY AS A MEMBER OF ŠKOLSKA KNJIGA CRO TEAM.

AS A MEMBER OF TEAM RAGUSA CRO, ORSAT KRALJ ACHIEVED 3RD PLACE AT 2025 EUROPEAN RCJ COMPETITION IN BARI IN RESCUE SIMULATION CATEGORY. HE HAS ALSO PARTICIPATED AT WORLD ROBOCUP JUNIOR 2023. COMPETITION IN BORDEAUX ALONGSIDE LUKA KOLARIĆ

AT THIS YEARS CROATIAN NATIONAL COMPETITON, TEAM HAS ACHIVED 2ND PLACE IN RCJ RESCUE SIMULATION TO QUALIFY FOR WORLD COMPETITION IN INCHEON